

Module 1: Introduction to Networking

Study Notes

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Q1. Definition and Importance of Networking

Definition

A **computer network** is a collection of two or more computers and devices interconnected by communication links (wired or wireless) to share resources, exchange data, and communicate with each other.

Importance of Networking

- **Resource Sharing:** Printers, storage, and software can be shared across devices, reducing cost.
- **Communication:** Email, video calls, instant messaging — all rely on networks.
- **Data Sharing:** Files and databases can be accessed from multiple locations.
- **Centralized Management:** Administrators can manage users and data from one location.
- **Cost Efficiency:** One internet connection or printer can serve many users.
- **Reliability:** Data can be replicated across systems, preventing single-point failure.

Q2. Major Components of a Computer Network

1. **End Devices (Nodes):** Computers, smartphones, tablets — devices that originate or consume data.
2. **Network Interface Card (NIC):** Hardware in each device that connects it to the network (has a unique MAC address).
3. **Transmission Media:** The physical path data travels — twisted pair cables, coaxial cables, fiber optics, or wireless (Wi-Fi, Bluetooth).
4. **Hub/Switch:** A *hub* broadcasts data to all devices; a *switch* is smarter and sends data only to the intended device using MAC addresses.
5. **Router:** Connects different networks and routes data packets between them using IP addresses. Essential for internet access.
6. **Modem:** Modulates/demodulates signals to convert digital data to analog (for telephone lines) and back.
7. **Protocols:** Rules that govern communication — e.g., TCP/IP, HTTP, FTP.
8. **Server:** A powerful computer that provides services (web, file, email) to client devices.

Q3. LAN, MAN, and WAN

Definition

Networks are classified by their **geographical coverage**:

- **LAN** (Local Area Network): Within a building or campus.
- **MAN** (Metropolitan Area Network): Spans a city.
- **WAN** (Wide Area Network): Spans countries or the globe.

Feature	LAN	MAN	WAN
Coverage	Building/Campus	City/Town	Country/World
Speed	Very High (1–10 Gbps)	Moderate (100 Mbps)	Low–Moderate (varies)
Ownership	Private	Public/Private	Public/ISPs
Cost	Low	Medium	High
Error Rate	Low	Moderate	Higher
Example	Office network	City cable TV	Internet

Q4. Any Two Network Topologies (Brief)

Topology refers to the physical or logical arrangement of nodes and links in a network.

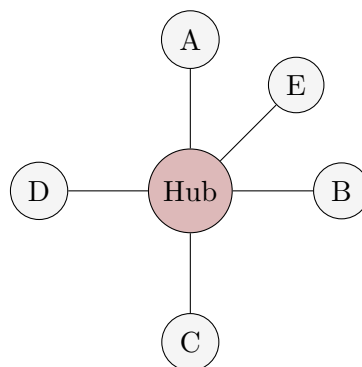
1. Star Topology: All devices connect to a central hub or switch. Easy to manage; failure of one device doesn't affect others; but if the hub fails, the whole network goes down.

2. Bus Topology: All devices share a single communication line (backbone). Simple and cheap; but a break in the cable disrupts the entire network and performance drops as more devices are added.

Q5. Star, Ring, Bus, and Mesh Topologies with Diagrams

5.1 Star Topology

All devices connect to a central hub/switch. Data passes through the hub.

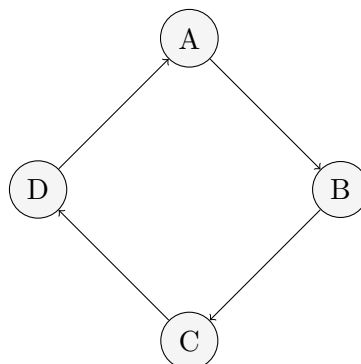


Pros: Easy troubleshooting; one device failure doesn't affect others.

Cons: Hub is a single point of failure; requires more cable than bus.

5.2 Ring Topology

Each device connects to exactly two others, forming a closed loop. Data travels in one direction (or both in dual ring).

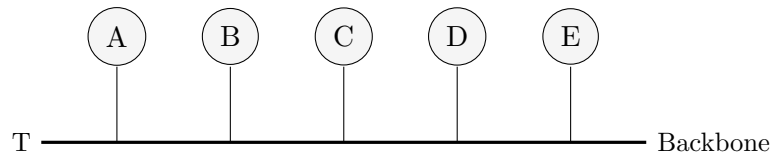


Pros: Equal access for all devices; performs well under heavy load.

Cons: A single cable break can disrupt the whole network.

5.3 Bus Topology

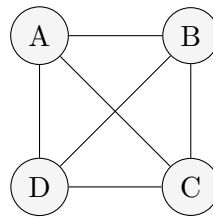
All devices share a single backbone cable. Each device taps into the bus.



Pros: Inexpensive; easy to install for small networks.
Cons: Collision-prone; backbone failure kills the whole network.

5.4 Mesh Topology

Every device connects directly to every other device. Provides maximum redundancy.



Pros: Highly reliable; no single point of failure; multiple paths.
Cons: Very expensive; complex wiring. For n devices: $\frac{n(n-1)}{2}$ links needed.

Q6. OSI Reference Model

Definition

The **OSI (Open Systems Interconnection)** model, developed by ISO in 1984, is a conceptual framework that standardizes how different network systems communicate. It has **7 layers**.

Layer #	Layer Name	Function
7	Application	Interface for user applications; handles HTTP, FTP, SMTP, DNS.
6	Presentation	Data translation, encryption, and compression (e.g., JPEG, SSL).
5	Session	Establishes, maintains, and terminates communication sessions.
4	Transport	End-to-end delivery, segmentation, flow control, error recovery (TCP/UDP).
3	Network	Logical addressing and routing of packets (IP, routers).
2	Data Link	Frame packaging, MAC addressing, error detection (switches, Ethernet).
1	Physical	Transmission of raw bits over physical media (cables, hubs, voltages).

Memory tip (top to bottom): *All People Seem To Need Data Processing*

Key Concepts Per Layer

- **Layer 7 (Application):** Closest to the user. Provides services like web browsing (HTTP), email (SMTP), file transfer (FTP).

- **Layer 6 (Presentation):** Acts as a translator. Converts formats (e.g., ASCII to EBCDIC), compresses data, encrypts/decrypts.
- **Layer 5 (Session):** Manages dialogue control — who talks when (half/full duplex). Handles login sessions.
- **Layer 4 (Transport):** TCP provides reliable, ordered delivery. UDP is faster but unreliable. Uses port numbers.
- **Layer 3 (Network):** Adds IP addresses. Routers operate here. Responsible for finding the best path.
- **Layer 2 (Data Link):** Divided into LLC and MAC sub-layers. Ethernet, Wi-Fi operate here. Detects errors via CRC.
- **Layer 1 (Physical):** Defines connectors, cables, signal types. Bits are converted to electrical/optical signals.

Q7. TCP/IP Reference Model

Definition

The **TCP/IP model** (also called the Internet Model) is the practical protocol suite used on the internet. It has **4 layers** and was developed by DARPA.

Layer #	Layer Name	Function & Protocols
4	Application	User-level protocols: HTTP, FTP, SMTP, DNS, Telnet.
3	Transport	End-to-end communication: TCP (reliable), UDP (unreliable).
2	Internet	Logical addressing and routing: IP, ICMP, ARP.
1	Network Access (Link)	Physical transmission and framing: Ethernet, Wi-Fi, drivers.

How it works

- **Application Layer:** User interacts via apps; data formatted as messages.
- **Transport Layer:** TCP breaks data into segments, numbers them, ensures delivery. UDP sends datagrams without confirmation.
- **Internet Layer:** Adds source and destination IP addresses. IP is connectionless and best-effort.
- **Network Access Layer:** Handles hardware addressing (MAC) and signal transmission on the local link.

Q8. Comparison: OSI vs TCP/IP Model

Feature	OSI Model	TCP/IP Model
Layers	7	4
Developed by	ISO (1984)	DARPA (1970s)
Purpose	Conceptual/reference model	Practical implementation
Session & Presentation	Separate layers (5 & 6)	Merged into Application layer
Transport protocols	Generic	Specifically TCP and UDP
Network layer name	Network	Internet
Reliability	Defined at Transport & Data Link	Primarily at Transport (TCP)
Usage	Teaching and design reference	Real-world internet communication
Protocol dependency	Protocol-independent	Built around TCP/IP protocols

Key takeaway: OSI is the *theoretical blueprint* — it explains *what* should happen at each layer. TCP/IP is the *working implementation* — it defines *how* the internet actually works. OSI's layers 5, 6, 7 collapse into TCP/IP's Application layer; OSI's layers 1–2 collapse into TCP/IP's Network Access layer.